GIF COMPRESSION TECHNIQUE/FORMAT

ASPECT	DETAILS
Full Form	Graphics Interchange Format
Introduced By	CompuServe in **1987** (GIF87a), updated in **1989** (GIF89a with animation + transparency)
Compression Algorithm	**LZW (Lempel–Ziv–Welch)** – dictionary-based, lossless
Compression Nature	**Lossless** (original data can be perfectly reconstructed, but limited colors reduce fidelity)
Color Model	Indexed color (uses a **Color Look-Up Table / Palette**)
Color Depth	Maximum **8-bit (256 colors per frame)**
Transparency Support	Yes (1-bit transparency → one palette entry can be marked transparent)
Interlacing	Supported (progressive display of image while loading in multiple passes)
Animation Support	Yes (multi-frame images → animated GIFs, controlled by frame delay & loop count)
Compression Efficiency	High for flat-colored images (logos, icons, cartoons); poor for photographic images
Data Structure	GIF file contains: **Header, Logical Screen Descriptor, Global Color Table, Image Descriptor(s), Local Color Table(s), Compressed Image Data (via LZW), Trailer**
Working Process	1. Convert image to indexed colors (max 256) 2. Store pixel values as indices 3. Apply LZW to compress index sequence 4. Store along with palette in GIF file
File Size	Smaller for simple/flat images; larger for detailed photos compared to JPEG/WebP (Web Picture Format)
Strengths	- Wide support across platforms - Animation capability - Transparency (basic) - Lossless compression for pixel data
Limitations	- Limited to 256 colors → poor for high-quality photos - Larger file sizes than JPEG/WebP for complex images - No full alpha transparency (only 1-bit), **Alpha** = the **fourth channel** in an image (after Red, Green, Blue).

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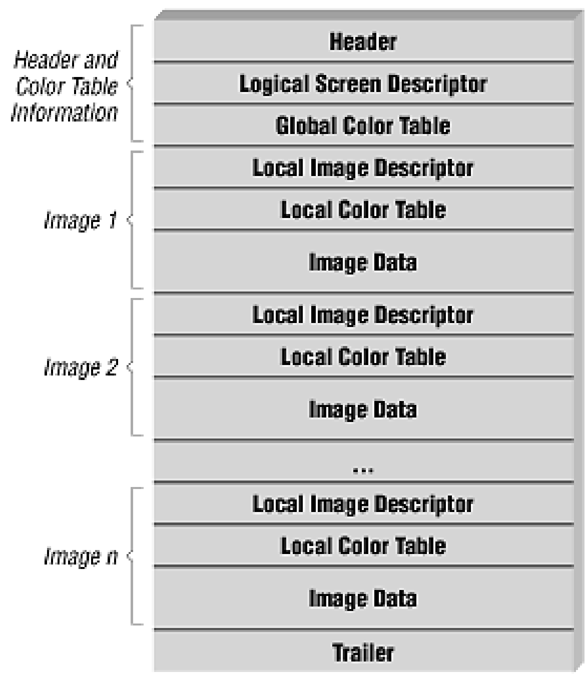
Best Use Cases	Memes, simple animations, icons, web logos, diagrams, low-color graphics
Not Suitable For	High-resolution photos, images requiring millions of colors, large modern animations
Modern Alternatives	- **PNG** (better for still images with transparency) - **JPEG** (better for photos) - **WebP/APNG** (better for animations + compression)

GIF only supports 1-bit transparency:

- Out of the **256 colors in the palette**, **only one color can be set as "transparent."**
- That means pixels using that palette entry will be invisible.
- Example: If color index #15 is marked as transparent, every pixel using that index won't be shown.



Header

The Header is six bytes in size and is used only to identify the file as type GIF.

Trailer

The Trailer is a single byte of data that occurs as the last character in the file. This byte value is always 3Bh and indicates the end of the GIF data stream. A trailer must appear in every GIF file.